the warps and curves on the floor's surface. Einstein applied this simple idea to the fabric of the universe. He imagined that in the absence of matter or energy—no sun, no earth, no stars—spacetime, like the smooth wooden floor, has no warps or curves. It's flat. This is schematically illustrated in Figure 3.9a, in which we focus on one slice of space. Of course, space is really three dimensional, and so Figure 3.9b is a more accurate depiction, but drawings that illustrate two dimensions are easier to understand, so we'll continue to use them. Einstein then imagined that the presence of matter or energy has an effect on space much like the effect the flood had on the floor. Matter and energy, like the sun, cause space (and spacetime[5]) to warp and curve as illustrated in Figures 3.10a and 3.10b. And just as a marble rolling on the warped floor travels along a curved path, Einstein showed that anything moving through warped space—such as the earth moving in the vicinity of the sun—will travel along a curved trajectory, as illustrated in Figure 3.11a and Figure 3.11b.

It's as if matter and energy imprint a network of chutes and valleys along which objects are guided by the invisible hand of the spacetime fabric. That, according to Einstein, is how gravity exerts its influence. The same idea also applies closer to home. Right now, your body would like to slide down an indentation in the spacetime fabric caused by the earth's presence. But your motion is being blocked by the surface on which you're sitting or standing. The upward push you feel almost every moment of your life—be it from the ground, the floor of your house, the corner easy chair, or your kingsize bed—is acting to stop you from sliding down a valley in spacetime. By contrast, should you throw yourself off the high diving board, you are giving in to gravity by allowing your body to move freely along one of its spacetime chutes.

